

[14X030] Introduction to Computational Finance

Exercise #1

February 24, 2026

General instructions

Each student is expected to upload on Moodle a **zip** file containing:

- A report in **pdf** format to answer exercise questions.
- The **code** used to generate the results of the report.
- If preferred, submit a unique **Jupyter notebook** with both code and report explanations.

Deeper instructions

- **Report:**
 - Answers to the TP questions. Figures and numerical results are necessary but **strictly not** sufficient: provide your analysis/comments as well.
 - Example: if your results surprise you, explain why and explain what the expected result was.
 - **Pay attention** to the presentation: axis labels, legend, etc.
- **Code:**
 - Implementation questions + code used to produce the results and figures in the report.
 - Programming language of your choice.
 - Do not use ChatGPT, or any other LLM/AI-assisted tools while writing down your code. Try by yourself, otherwise it will be counted as failed.
 - I recommend Python/Julia: I won't be able to assist you in the same way if you use a different language.

Deadline

Any questions to: [Lorenzo Bini](#).

Upload on Moodle due by : **March 2, 2026 at 11:59 pm.**

1 Computing returns

The following series describes the value of a portfolio at the end of each month:

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
t	0	1	2	3	4	5	6	7	8	9	10	11	12
P(t)	100	101	102	103	104	105	106	107	108	109	110	111	112

The initial value of the portfolio being 100, the annual return is :

$$R_a = R_{12}(12) = \frac{P(12) - P(0)}{P(0)} = 12\%$$

1. Compute monthly returns $\{R(t) \mid t = 1, \dots, 12\}$.
2. Compute the annual return R_a from the monthly returns (you should find 12% again). Compare it to the sum of the monthly returns.
3. Compute the average monthly return R_m and compare it to the average of the monthly returns.

2 Continuously compounded return

Consider a constant annual interest rate of $R = 2.11\%$ with a continuous compounding.

1. With a initial amount of \$10000, what will be the value in:
 - (a) 1 year?
 - (b) 5 years?
 - (c) 10 years?
2. What is the initial amount to have a final amount of \$10000 in:
 - (a) 1 year?
 - (b) 3 years?
 - (c) 10 years?

3 Portfolio of Microsoft and Starbucks stock

You purchase 10 shares of each Microsoft and Starbucks stock at the end of month $t - 1$ at prices $P_{msft}(t - 1) = \$85$ and $P_{sbux}(t - 1) = \$30$.

1. Compute $V(t - 1)$, the initial value of the portfolio.
2. Compute the portfolio shares α_{msft} and α_{sbux} .

Consider now that at the end of month t , the prices are $P_{msft}(t) = \$90$ and $P_{sbux}(t) = \$28$.

3. Compute $R_{msft}(t)$ and $R_{sbux}(t)$, the one-period return of Microsoft and Starbucks stocks.
4. Compute the one-period return of the portfolio $R(t)$ and its value $V(t)$ at the end of month t .

4 Present / Future value

Consider a constant annual return $R = 4.5\%$, semi-annually compounded (*i.e.* two payments a year).

1. With an initial amount of \$10000, what will be the value in:
 - (a) 1 year?
 - (b) 5 years?
 - (c) 10 years?
2. What is the initial amount to have a final amount of \$10000 in:
 - (a) 1 year?
 - (b) 3 years?
 - (c) 10 years?
3. Optional : Let R be the one-period return, n the number of periods, and m the number of payments per period. How would you prove that :

$$\lim_{m \rightarrow +\infty} \left(1 + \frac{R}{m}\right)^{mn} = e^{Rn}.$$